11.2.5 Nano in laboratories

The nanostructures made in the laboratory mimic some of nature's amazing nanostructures. As the nanostructures are so small, specialized methods are needed to manufacture objects in this size range. There are two ways of preparing the nanomaterials, top down and bottom up approaches.

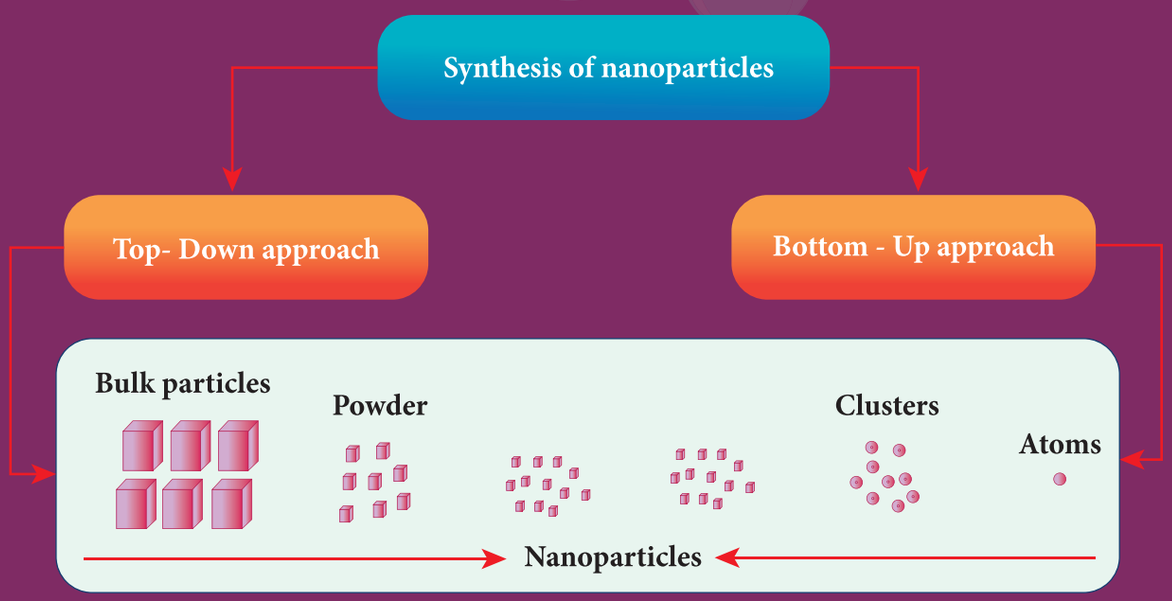
Synthesis of nanoparticles

Top Down Approach

Nanomaterials are synthesised by breaking down bulk solids into nano sizes. Example, Ball milling, sol-gel, lithography

Bottom Up Approach

Nanomaterials are synthesised by assembling the atoms/molecules together. Selectively atoms are added to create structures. Example, plasma etching, and chemical vapour deposition



Due to the vast available methods the details are avoided at this level.

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